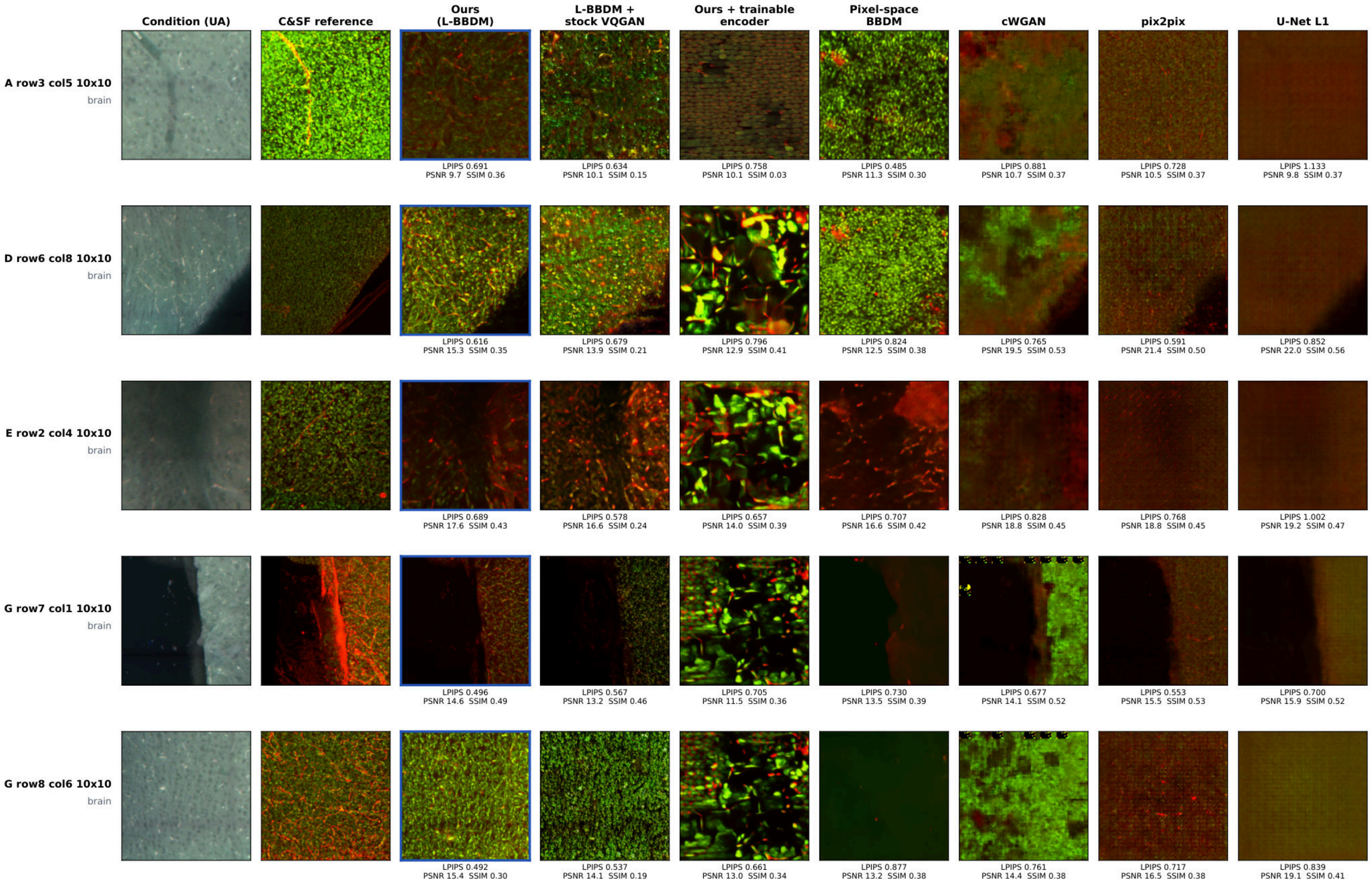
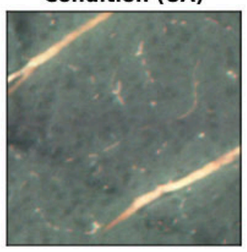
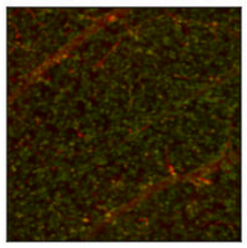
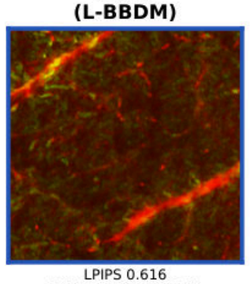
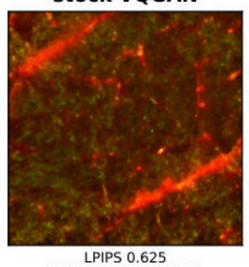
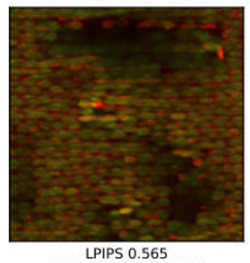
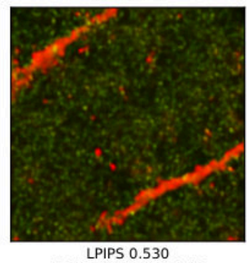
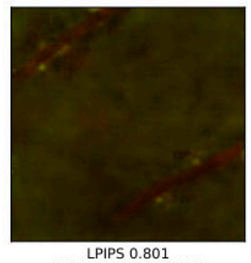
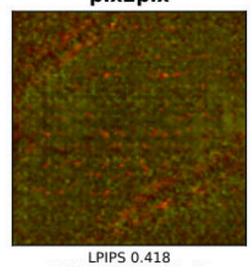
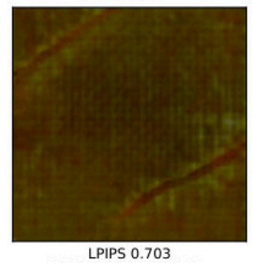
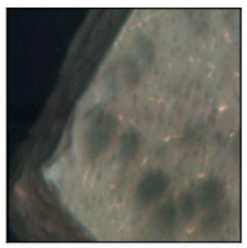
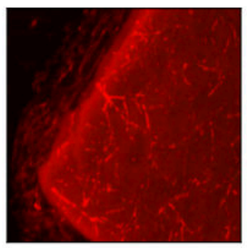
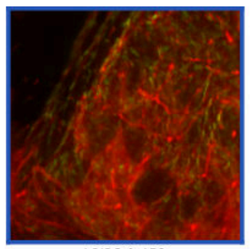
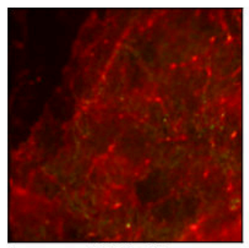
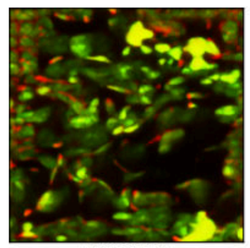
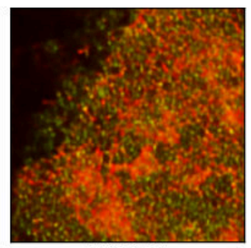
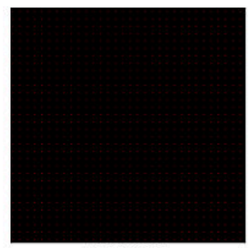
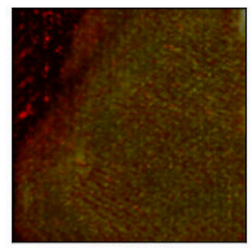
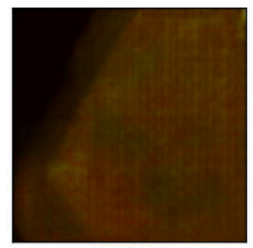
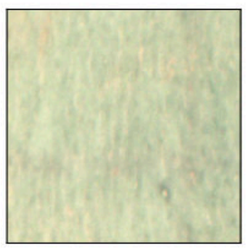
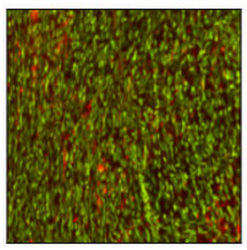
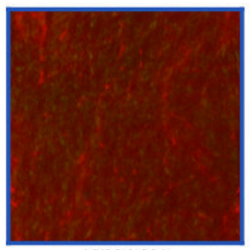
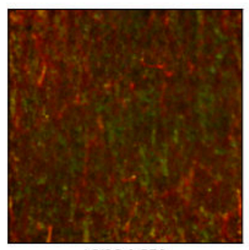
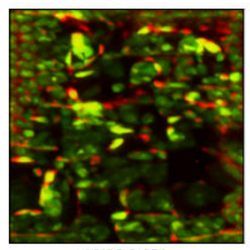
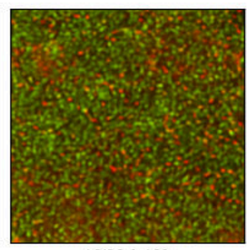
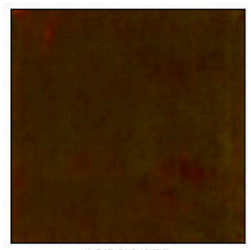
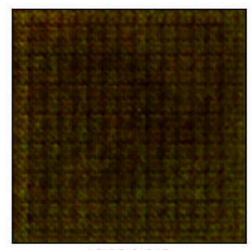
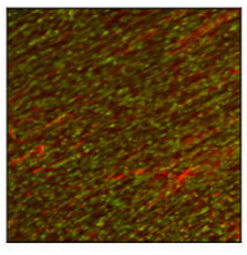
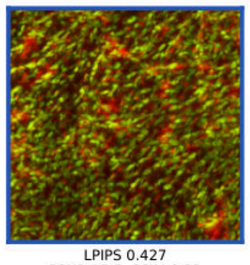
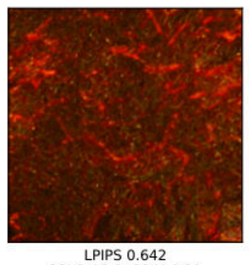
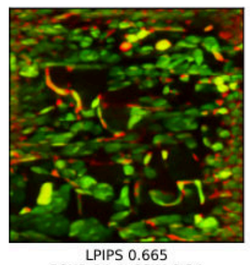
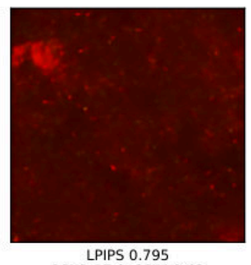
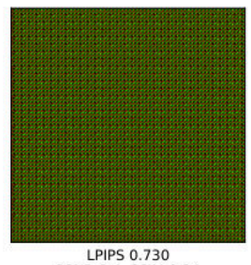
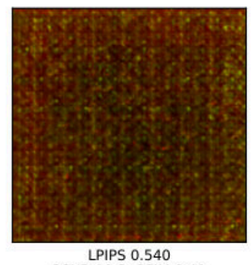
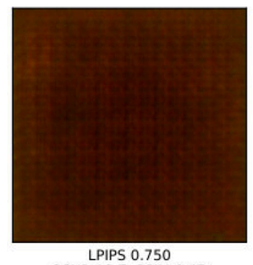
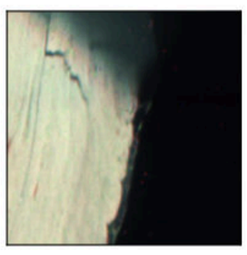
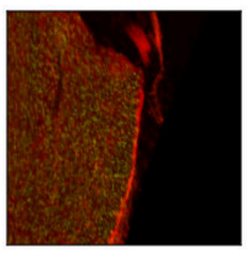
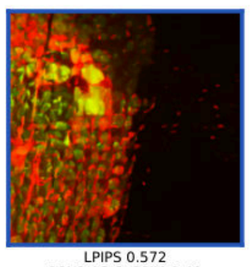
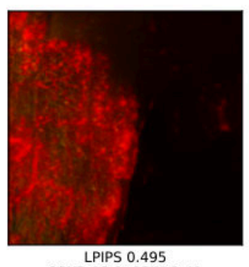
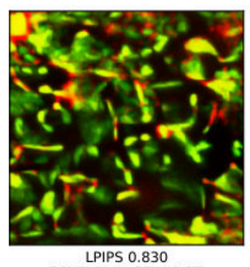
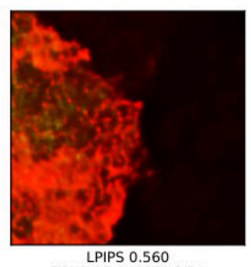
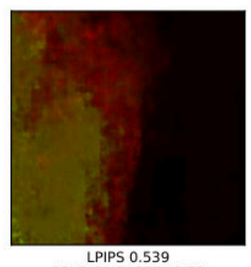
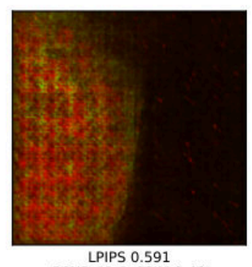
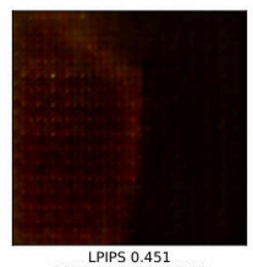
General tissue texture: Leave-one-specimen-out (piece never seen in training) (1/3)



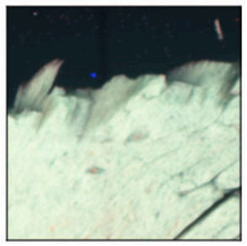
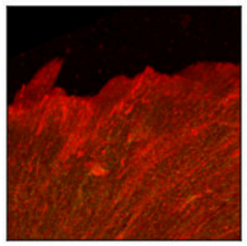
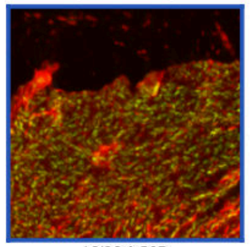
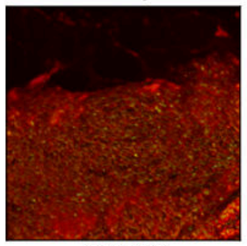
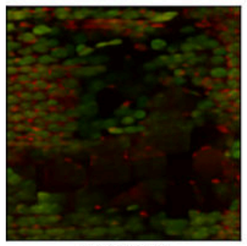
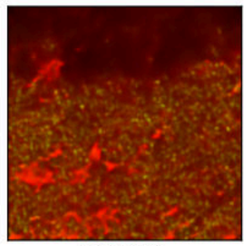
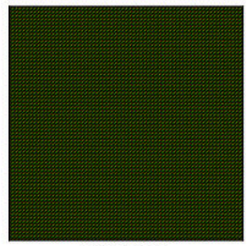
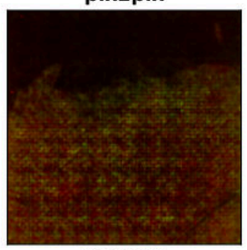
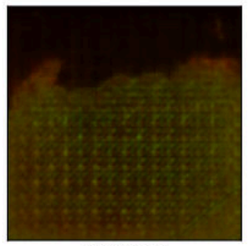
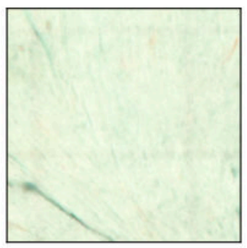
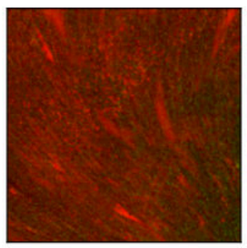
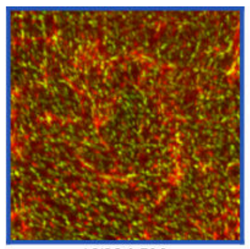
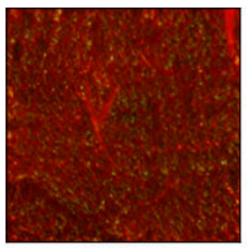
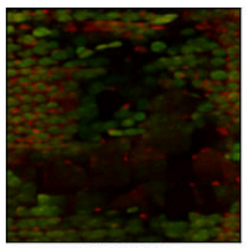
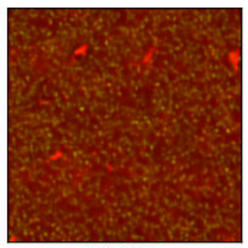
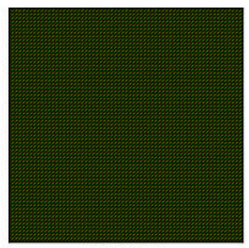
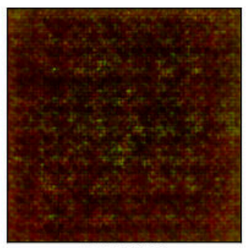
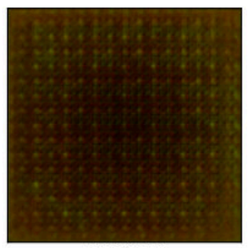
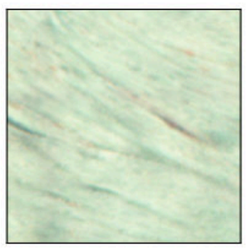
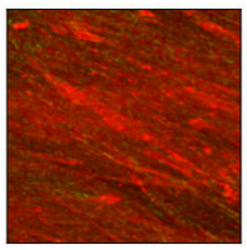
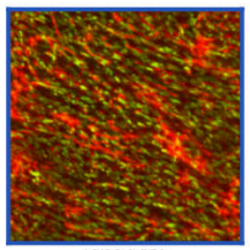
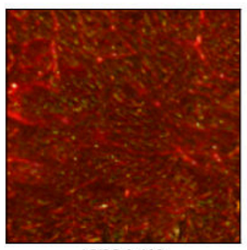
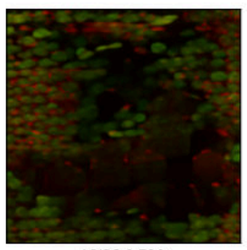
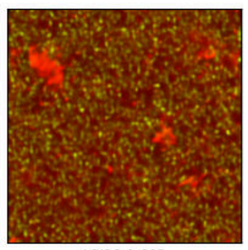
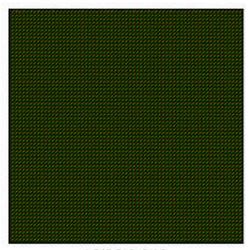
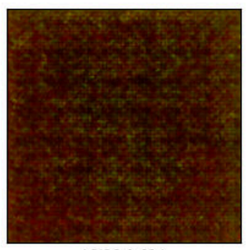
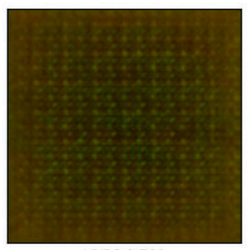
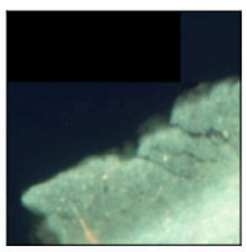
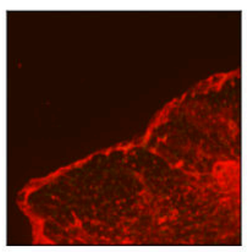
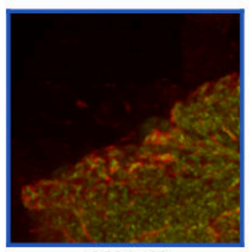
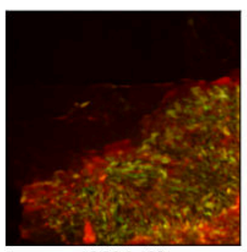
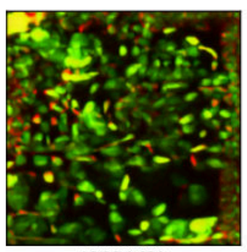
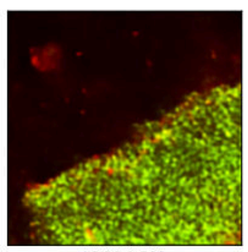
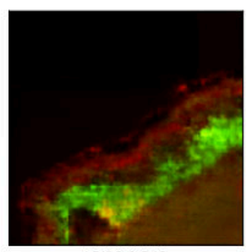
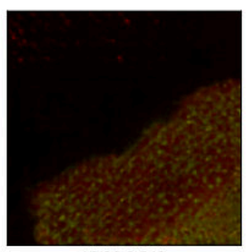
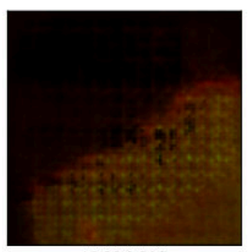
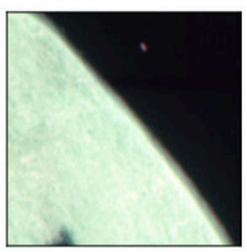
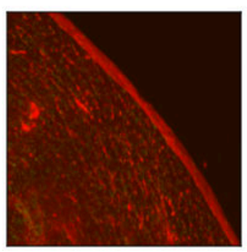
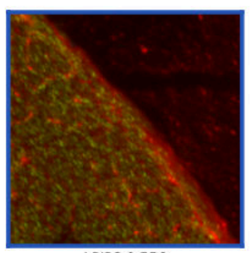
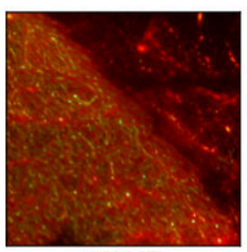
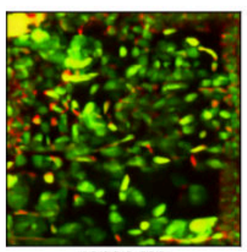
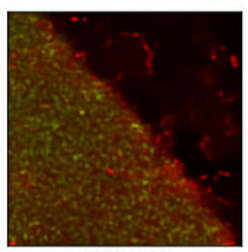
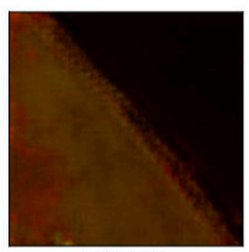
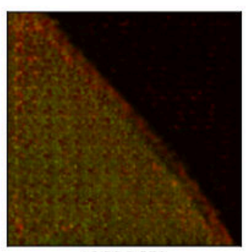
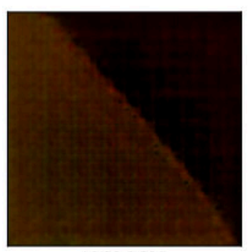
Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

General tissue texture: Leave-one-specimen-out (piece never seen in training) (2/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix	U-Net L1
H row2 col8 10x10 brain			 LPIPS 0.616 PSNR 18.1 SSIM 0.45	 LPIPS 0.625 PSNR 16.6 SSIM 0.28	 LPIPS 0.565 PSNR 19.9 SSIM 0.18	 LPIPS 0.530 PSNR 19.3 SSIM 0.48	 LPIPS 0.801 PSNR 24.6 SSIM 0.56	 LPIPS 0.418 PSNR 19.0 SSIM 0.46	 LPIPS 0.703 PSNR 21.2 SSIM 0.52
I row7 col0 10x10 brain			 LPIPS 0.452 PSNR 18.8 SSIM 0.48	 LPIPS 0.435 PSNR 19.6 SSIM 0.59	 LPIPS 0.724 PSNR 11.8 SSIM 0.42	 LPIPS 0.731 PSNR 15.5 SSIM 0.43	 LPIPS 0.895 PSNR 12.0 SSIM 0.43	 LPIPS 0.682 PSNR 16.8 SSIM 0.48	 LPIPS 0.741 PSNR 17.4 SSIM 0.56
B row6 col3 10x10 heart			 LPIPS 0.804 PSNR 15.2 SSIM 0.38	 LPIPS 0.570 PSNR 16.2 SSIM 0.31	 LPIPS 0.651 PSNR 13.7 SSIM 0.34	 LPIPS 0.460 PSNR 16.8 SSIM 0.37	 LPIPS 0.952 PSNR 15.9 SSIM 0.40	 LPIPS 0.643 PSNR 15.1 SSIM 0.39	 LPIPS 0.830 PSNR 13.3 SSIM 0.38
C row3 col4 10x10 heart			 LPIPS 0.427 PSNR 15.5 SSIM 0.33	 LPIPS 0.642 PSNR 18.0 SSIM 0.21	 LPIPS 0.665 PSNR 14.4 SSIM 0.34	 LPIPS 0.795 PSNR 17.0 SSIM 0.43	 LPIPS 0.730 PSNR 8.4 SSIM 0.34	 LPIPS 0.540 PSNR 19.2 SSIM 0.41	 LPIPS 0.750 PSNR 18.7 SSIM 0.45
F row3 col7 10x10 heart			 LPIPS 0.572 PSNR 15.6 SSIM 0.40	 LPIPS 0.495 PSNR 18.9 SSIM 0.41	 LPIPS 0.830 PSNR 11.1 SSIM 0.37	 LPIPS 0.560 PSNR 15.8 SSIM 0.54	 LPIPS 0.539 PSNR 21.2 SSIM 0.57	 LPIPS 0.591 PSNR 19.6 SSIM 0.42	 LPIPS 0.451 PSNR 19.0 SSIM 0.47

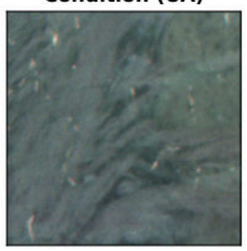
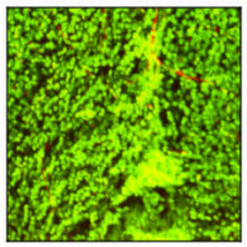
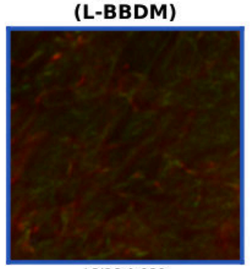
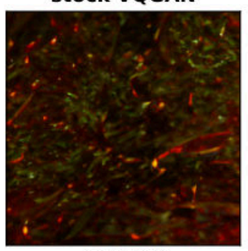
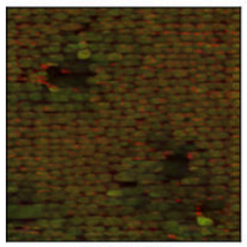
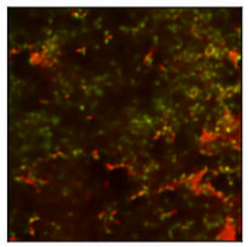
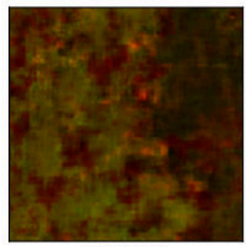
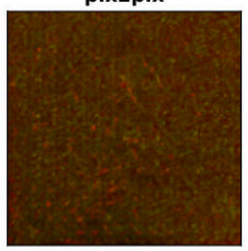
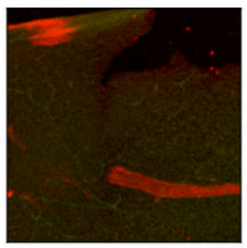
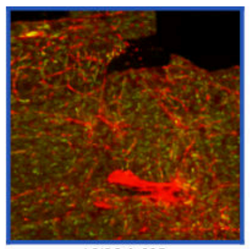
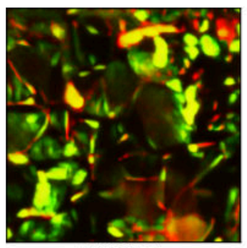
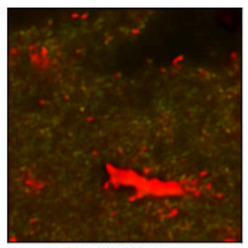
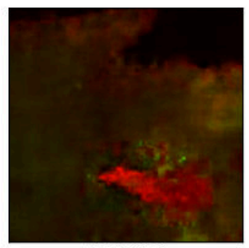
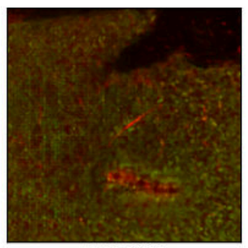
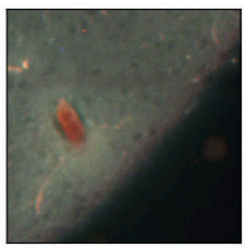
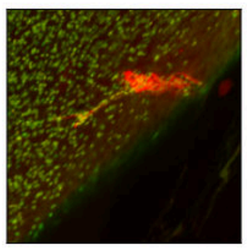
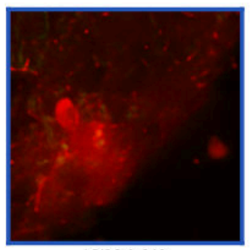
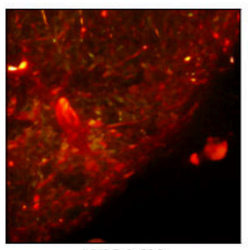
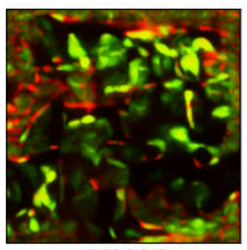
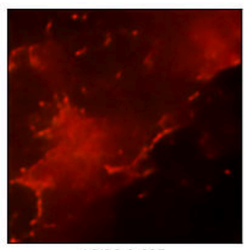
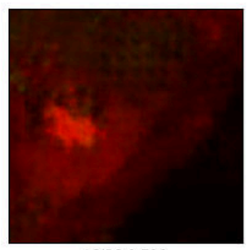
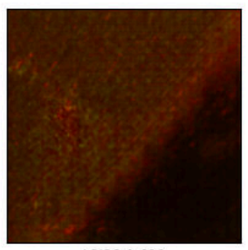
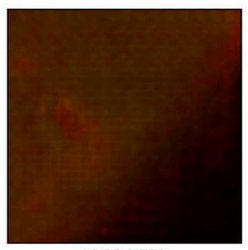
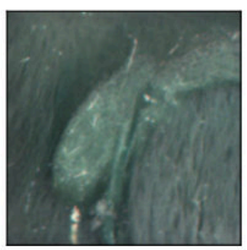
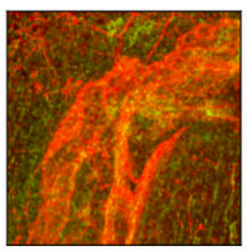
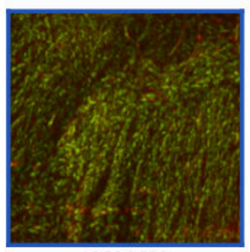
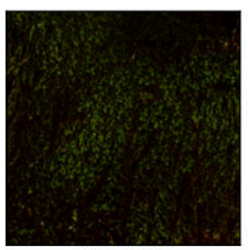
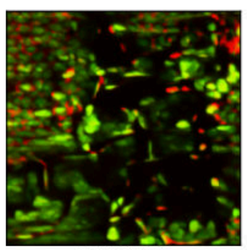
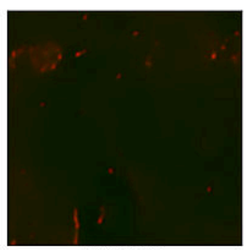
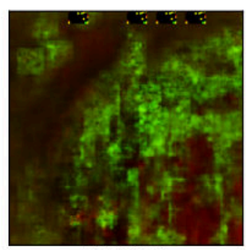
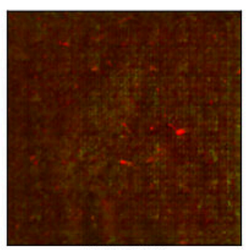
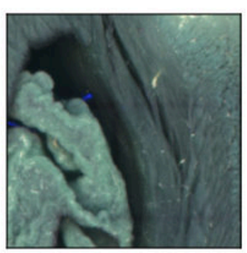
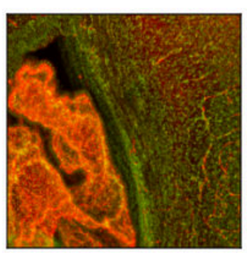
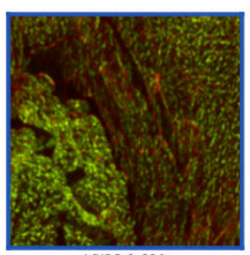
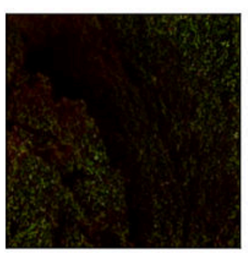
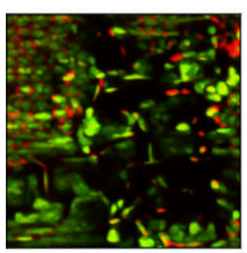
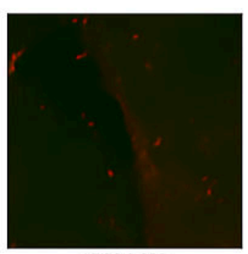
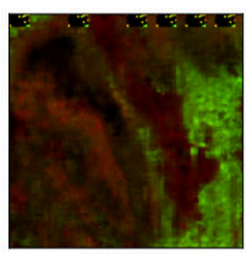
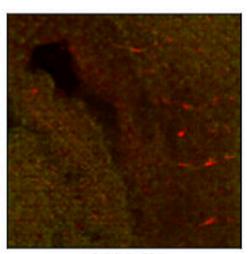
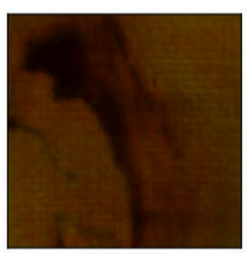
Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

General tissue texture: Leave-one-specimen-out (piece never seen in training) (3/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix	U-Net L1
J row0 col4 5x5 heart			 LPIPS 0.587 PSNR 17.6 SSIM 0.45	 LPIPS 0.488 PSNR 19.8 SSIM 0.40	 LPIPS 0.869 PSNR 13.6 SSIM 0.38	 LPIPS 0.695 PSNR 18.0 SSIM 0.42	 LPIPS 0.882 PSNR 8.9 SSIM 0.34	 LPIPS 0.610 PSNR 17.0 SSIM 0.51	 LPIPS 0.723 PSNR 16.5 SSIM 0.52
J row1 col3 5x5 heart			 LPIPS 0.726 PSNR 16.0 SSIM 0.28	 LPIPS 0.517 PSNR 21.2 SSIM 0.35	 LPIPS 0.880 PSNR 14.6 SSIM 0.38	 LPIPS 0.692 PSNR 21.3 SSIM 0.47	 LPIPS 0.877 PSNR 9.3 SSIM 0.34	 LPIPS 0.608 PSNR 17.2 SSIM 0.41	 LPIPS 0.718 PSNR 17.5 SSIM 0.49
J row5 col10 10x10 heart			 LPIPS 0.571 PSNR 15.7 SSIM 0.31	 LPIPS 0.492 PSNR 18.7 SSIM 0.30	 LPIPS 0.720 PSNR 12.3 SSIM 0.37	 LPIPS 0.693 PSNR 18.3 SSIM 0.40	 LPIPS 0.940 PSNR 8.5 SSIM 0.34	 LPIPS 0.634 PSNR 15.1 SSIM 0.43	 LPIPS 0.701 PSNR 15.0 SSIM 0.48
K row1 col1 10x10 heart			 LPIPS 0.488 PSNR 18.6 SSIM 0.51	 LPIPS 0.488 PSNR 18.3 SSIM 0.46	 LPIPS 0.820 PSNR 11.9 SSIM 0.38	 LPIPS 0.588 PSNR 13.3 SSIM 0.34	 LPIPS 0.543 PSNR 16.4 SSIM 0.45	 LPIPS 0.543 PSNR 17.9 SSIM 0.46	 LPIPS 0.596 PSNR 18.9 SSIM 0.53
K row3 col6 10x10 heart			 LPIPS 0.556 PSNR 19.5 SSIM 0.48	 LPIPS 0.578 PSNR 17.8 SSIM 0.41	 LPIPS 0.836 PSNR 12.5 SSIM 0.38	 LPIPS 0.636 PSNR 19.5 SSIM 0.48	 LPIPS 0.612 PSNR 19.8 SSIM 0.52	 LPIPS 0.576 PSNR 18.9 SSIM 0.49	 LPIPS 0.590 PSNR 19.0 SSIM 0.53

Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

Biological features: Leave-one-specimen-out (piece never seen in training) (1/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix	U-Net L1
A row7 col5 10x10 brain									
			LPIPS 0.832 PSNR 8.4 SSIM 0.35	LPIPS 0.732 PSNR 8.5 SSIM 0.19	LPIPS 0.743 PSNR 9.9 SSIM 0.03	LPIPS 0.695 PSNR 8.7 SSIM 0.23	LPIPS 0.930 PSNR 9.6 SSIM 0.37	LPIPS 0.807 PSNR 9.8 SSIM 0.37	LPIPS 1.133 PSNR 9.2 SSIM 0.37
D (Zpart1) row1 col2 5x5 brain									
			LPIPS 0.625 PSNR 16.1 SSIM 0.40	LPIPS 0.722 PSNR 12.1 SSIM 0.18	LPIPS 0.771 PSNR 12.2 SSIM 0.43	LPIPS 0.637 PSNR 20.5 SSIM 0.60	LPIPS 0.658 PSNR 21.9 SSIM 0.66	LPIPS 0.627 PSNR 20.7 SSIM 0.56	LPIPS 0.670 PSNR 22.2 SSIM 0.69
E row6 col10 10x10 brain									
			LPIPS 0.649 PSNR 17.5 SSIM 0.58	LPIPS 0.629 PSNR 16.9 SSIM 0.42	LPIPS 0.646 PSNR 13.7 SSIM 0.45	LPIPS 0.635 PSNR 17.3 SSIM 0.50	LPIPS 0.720 PSNR 17.7 SSIM 0.54	LPIPS 0.629 PSNR 20.0 SSIM 0.58	LPIPS 0.779 PSNR 20.2 SSIM 0.60
G row1 col1 10x10 brain									
			LPIPS 0.596 PSNR 12.1 SSIM 0.35	LPIPS 0.792 PSNR 9.2 SSIM 0.27	LPIPS 0.733 PSNR 9.7 SSIM 0.34	LPIPS 0.884 PSNR 9.4 SSIM 0.38	LPIPS 0.718 PSNR 11.2 SSIM 0.40	LPIPS 0.700 PSNR 12.2 SSIM 0.40	LPIPS 0.818 PSNR 13.3 SSIM 0.43
G row3 col2 5x5 brain									
			LPIPS 0.621 PSNR 14.3 SSIM 0.32	LPIPS 0.676 PSNR 10.9 SSIM 0.27	LPIPS 0.840 PSNR 11.4 SSIM 0.34	LPIPS 0.889 PSNR 11.0 SSIM 0.37	LPIPS 0.776 PSNR 12.5 SSIM 0.40	LPIPS 0.680 PSNR 13.7 SSIM 0.41	LPIPS 0.844 PSNR 14.5 SSIM 0.43

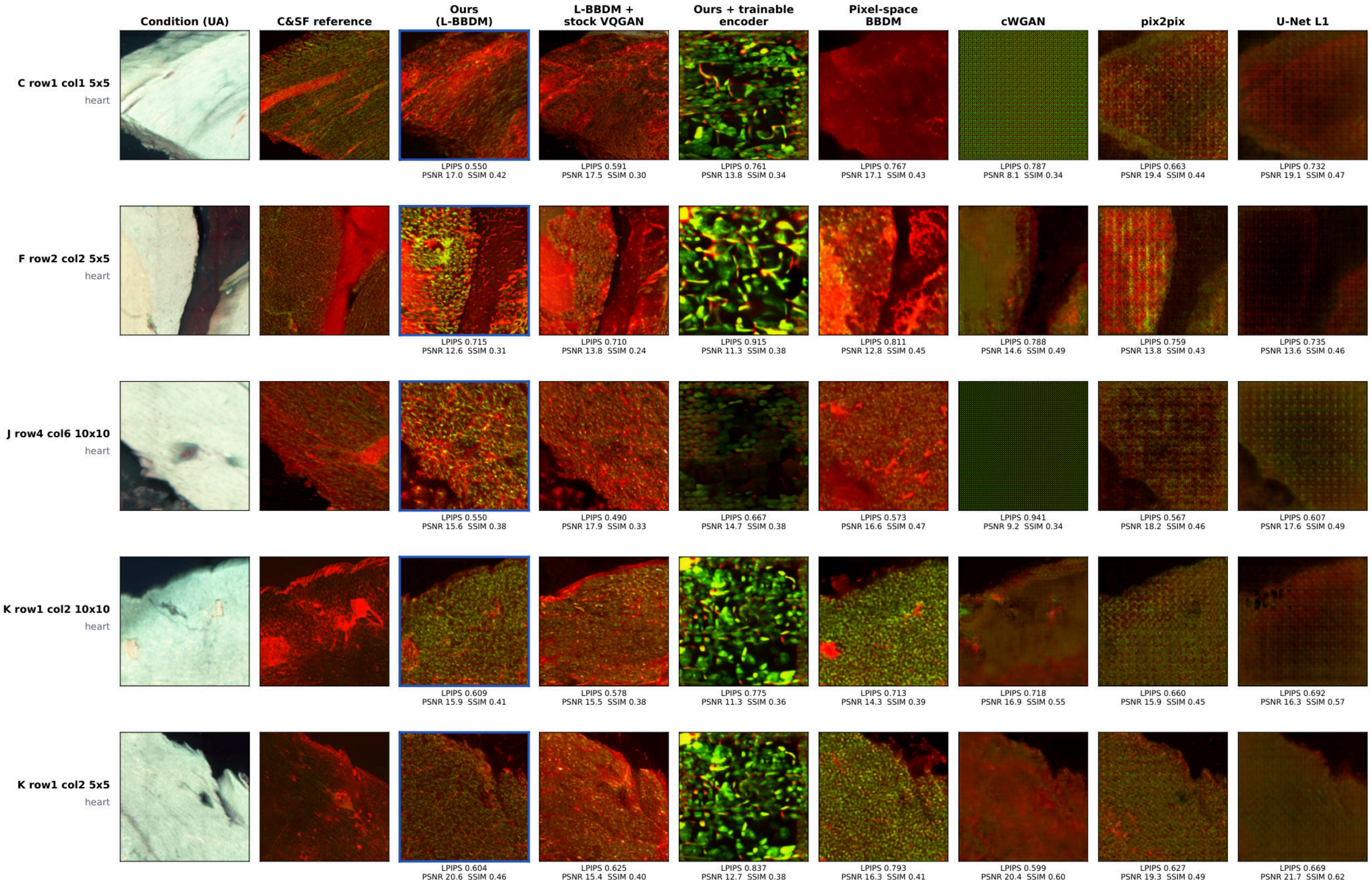
Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

Biological features: Leave-one-specimen-out (piece never seen in training) (2/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix	U-Net L1
G row4 col5 10x10 brain			LPIPS 0.448 PSNR 14.9 SSIM 0.34	LPIPS 0.409 PSNR 15.2 SSIM 0.17	LPIPS 0.639 PSNR 14.0 SSIM 0.35	LPIPS 0.742 PSNR 16.0 SSIM 0.41	LPIPS 0.686 PSNR 14.5 SSIM 0.40	LPIPS 0.777 PSNR 17.1 SSIM 0.41	LPIPS 0.827 PSNR 18.1 SSIM 0.43
G row5 col6 10x10 brain			LPIPS 0.531 PSNR 15.0 SSIM 0.41	LPIPS 0.535 PSNR 13.0 SSIM 0.24	LPIPS 0.729 PSNR 11.7 SSIM 0.35	LPIPS 0.750 PSNR 13.8 SSIM 0.39	LPIPS 0.701 PSNR 14.8 SSIM 0.44	LPIPS 0.570 PSNR 16.1 SSIM 0.44	LPIPS 0.776 PSNR 16.8 SSIM 0.46
H row4 col6 10x10 brain			LPIPS 0.694 PSNR 16.4 SSIM 0.46	LPIPS 0.556 PSNR 16.8 SSIM 0.27	LPIPS 0.607 PSNR 17.2 SSIM 0.19	LPIPS 0.506 PSNR 17.1 SSIM 0.45	LPIPS 0.743 PSNR 17.8 SSIM 0.55	LPIPS 0.567 PSNR 16.6 SSIM 0.45	LPIPS 0.753 PSNR 17.5 SSIM 0.53
I row3 col3 10x10 brain			LPIPS 0.518 PSNR 19.7 SSIM 0.50	LPIPS 0.461 PSNR 20.9 SSIM 0.59	LPIPS 0.723 PSNR 13.0 SSIM 0.43	LPIPS 0.393 PSNR 20.7 SSIM 0.63	LPIPS 0.843 PSNR 14.8 SSIM 0.36	LPIPS 0.776 PSNR 16.3 SSIM 0.37	LPIPS 0.710 PSNR 19.9 SSIM 0.58
B row7 col1 10x10 heart			LPIPS 0.815 PSNR 13.8 SSIM 0.37	LPIPS 0.597 PSNR 14.3 SSIM 0.29	LPIPS 0.651 PSNR 11.5 SSIM 0.33	LPIPS 0.516 PSNR 14.2 SSIM 0.37	LPIPS 0.944 PSNR 13.6 SSIM 0.39	LPIPS 0.647 PSNR 13.3 SSIM 0.37	LPIPS 0.857 PSNR 11.0 SSIM 0.37

Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

Biological features: Leave-one-specimen-out (piece never seen in training) (3/3)



Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.